

Full-Stack Software Engineer & Mobile Developer and NIT Rourkela CSE Graduate with 3+ years of experience engineering scalable web systems and cross-platform mobile apps (React Native / Expo). \$1M Bloomberg Global Challenge Winner with 10+ shipped production applications serving 100,000+ active users across TypeScript, Next.js, React, Fastify, and PostgreSQL.

KEY ACHIEVEMENTS & PRESTIGIOUS AWARDS

- Winner — Bloomberg Philanthropies Global Mayors Challenge 2021

18 January 2022

[Link](#)

Bloomberg Philanthropies

- Key engineering contributor to the winning \$1 Million international project for developing solar-powered, IoT-monitored cold storage solutions to eliminate post-harvest agricultural losses in Rourkela, India.
- Indian Council for Cultural Relations (ICCR) Scholarship

July 2018 - June 2022

Government of India

- Awarded a highly competitive, fully funded scholarship by the Government of India to pursue a 4-year B.Tech in Computer Science and Engineering at NIT Rourkela, representing Bangladesh as an international scholar.

SKILLS

Tools and Languages	JavaScript, TypeScript, Python, C, Git, AI
Frameworks & Databases	React, React Native, Next.js, Three.js, Redux, Zustand, Node.js, Fastify, Express.js, MongoDB, Mongoose, PostgreSQL, MySQL, HTML5, CSS3, Tailwind CSS, Docker, Firebase
State, Architecture & AI	Zustand, Redux Toolkit, Multi-Tenant Architecture, Role-Based Access Control (RBAC), Mobile Performance Optimization, REST APIs, GraphQL, WebSockets, Pusher, AI Text-to-Speech (ElevenLabs, Azure Speech TTS), Better Auth, JWT, OAuth
Relevant Courses	Data Structures & Algorithms, Database Management Systems (DBMS), Software Engineering, Software Project Process & Quality Management, Software Testing, Business Research Methodology
Communications	Bengali (Native), English, Hindi

TECHNICAL EXPERIENCE

- Daffodil International University (DIU)

December 2024 — Present

Software Engineer

Dhanmondi, Dhaka

- Architect and maintain production React Native & Expo mobile apps (iOS & Android) serving 100,000+ active learners across university e-learning and healthcare initiatives.
 - Build modular UI component libraries, custom hooks, and centralized client state management using Zustand and Redux Toolkit.
 - Integrate core services: REST APIs, OAuth/JWT authentication, local payment gateways, video streaming, and push notifications (FCM).
 - Profile mobile runtime performance (sustaining fluid 60 FPS) and manage end-to-end releases on Google Play Console and Apple App Store.
- Techdojo Limited

June 2023 — November 2024

Software Engineer

Uttara, Dhaka

- Delivered full-stack web platforms, mobile apps, and interactive 3D solutions using the MERN Stack, React Native, and Three.js.
 - Engineered a 3D VR Classroom Simulation in Three.js/WebGL, integrating ElevenLabs & Azure TTS for dynamic AI voice narration across lesson stages.
 - Designed secure RESTful microservices with Zod validation, rate limiting, and optimized MongoDB aggregation pipelines to minimize API latency.
 - Enforced clean code standards, conducted peer reviews, and integrated external APIs, payment gateways.
- Koel Fresh Private Limited

May 2021 — August 2021

Developer Associate Intern

Rourkela, India

- Built the IoT telemetry web dashboard in React as a core contributor to the \$1M Bloomberg Global Mayors Challenge 2021 winning project.
 - Developed real-time sensor monitoring dashboards visualizing live temperature, battery voltage, and solar power generation.
 - Integrated hardware sensor streams with cloud REST APIs, implementing client-side polling and caching for low-bandwidth rural networks.

PROJECTS

Rx Studio Doctor OS — Multi-Tenant Clinical SaaS Platform

2026

Technologies: Next.js, React, TypeScript, Fastify, Drizzle ORM, PostgreSQL, Docker, Better Auth, Tailwind CSS, Radix UI

[Rx-Studio](#)

- Architected a multi-tenant clinical SaaS with Fastify and Next.js, featuring isolated PostgreSQL database schemas, tenant slug routing, and Docker containerization.
- Built a dynamic prescription engine generating high-resolution PDF prescriptions with QR verification, custom pad margins, and digital doctor signatures.
- Engineered real-time OPD patient queues and chamber schedules via WebSockets, alongside Electronic Health Records (EHR) and drug interaction catalogs.

GoEdu Mobile Application — Leading Bangladesh EdTech Platform

2026

Technologies: Expo, React Native, TypeScript, Zustand, REST APIs, Payment Gateways, FCM

[GoEdu](#)

- Delivered the primary React Native mobile client for GoEdu (Daffodil Family), serving 100,000+ registered learners with smooth 60 FPS video lesson streaming.
- Integrated local payment gateways for automated checkout, instant course enrollment, and digital invoice generation.
- Automated learning streak reminders, push notifications, and course recommendations via Firebase Cloud Messaging (FCM) to maximize student retention.

CPCN Mobile Application — Child Health & Nutrition Certification

2024-2025

Technologies: React Native, Expo, TypeScript, Zustand, REST APIs, Android & iOS

[CPCN](#)

- Built the official cross-platform medical e-learning app (React Native) for Daffodil International University, serving healthcare practitioners with offline-first data sync.
- Developed a 6-module video curriculum player with downloadable clinical materials, automated quiz evaluations, and instant verifiable digital certificate generation.

International Online University (IOU) Mobile Application

2024-2025

Technologies: Expo, React Native, TypeScript, Zustand, REST APIs, JWT, Push Notifications, Fastlane(Automation)

[IOU](#)

- Engineered the university mobile LMS client in React Native and Expo for semester course enrollment, syllabus access, and lecture archive streaming.
- Implemented secure JWT session management, student grade dashboards, assignment deadline alerts, and verified digital diploma downloads.

Skill Jobs & Skill Jobs Recruiter Mobile Applications

2025

Technologies: React-Native, TypeScript, Redux, REST APIs, Push Notification, Android, iOS

[SkillJobs](#) [Recruiter App](#)

- Developed cross-platform mobile apps for job seekers and recruiters using React Native and Redux Toolkit, supporting multi-criteria job search, digital resume building, and instant applications.
- Built recruiter-side talent management dashboards featuring vacancy posting, applicant tracking pipelines, AI candidate evaluation scoring, and interview scheduling workflows.
- Optimized large job feed virtualization and cached API responses to sustain smooth 60 FPS scrolling and reduced mobile data consumption.

Global Education Axis CRM — Student Placement & Agency Platform

2025-2026

Technologies: Next.js, TypeScript, MongoDB, Better Auth, Pusher WebSockets, Tailwind CSS, DnD Kit, Cloudinary

- Architected an agency consultancy CRM in Next.js and MongoDB to manage international student recruitment, university application cycles, and partner commissions.
- Developed an interactive drag-and-drop Kanban candidate pipeline with DnD Kit to track applicants across admission, documentation, and visa stages.
- Engineered real-time counselor-student messaging using Pusher WebSockets and built a secure Cloudinary document vault for academic transcript verification.

Interactive 3D VR Classroom Simulation — Virtual Learning Platform

2023-2024

Technologies: Three.js, React, WebGL, React.js, JavaScript, ElevenLabs API, Azure Speech TTS

- Build an interactive in-browser 3D virtual classroom using Three.js and WebGL, featuring spatial 3D environments, lighting, camera controls, and 60 FPS rendering.
- Developed multi-phase classroom activity state machines integrated with ElevenLabs and Azure Speech TTS for real-time synthetic voice narration across lesson stages.

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Full-Stack MERN Application with JWT-Based Authentication and Security Features

September 2024

Technologies: React, Express, Zustand, MongoDB, Mongoose, JWT, Nodemailer, Shadcn/UI, Tailwind CSS

[GitHub](#) [Live-Website](#)

- Developed a production-ready authentication microservice featuring double-token rotation (short-lived access JWT + revolving HTTP-only refresh tokens) and CSRF protection.
- Built automated transactional email verification and password recovery with Nodemailer, backed by Zod input validation schemas.

Medical Shop Management System

April 2021 - Jun 2021

Technologies: Python, MySQL, Database Design, ER Diagrams, SDLC

- Collaborated on the development of a comprehensive Medical Shop Management System as part of a team project, following the Software Development Life Cycle (SDLC) methodology.
- Designed to streamline operations for medical stores, the system effectively managed inventory, stock, and accounting processes. Leveraged skills in database design, including ER Diagrams, and utilized Python and MySQL for implementation.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2018-2022

National Institute of Technology Rourkela, India

CGPA : 7.30/10 (First Class), Percentage : 73%

Scholarship: Full 4-year Government of India ICCR Scholarship recipient.

Higher Secondary Certificate

2014-2015

Birshreshtha Noor Mohammad Public College, Dhaka

GPA : 5.00/5.00

Secondary School Certificate

2012-2013

Ibn Taimiya School & College, Cumilla

GPA : 5.00/5.00

EXTRA CURRICULUM ACTIVITIES

Central Vigilance Commission Govt. of India

[Volunteer] August 2018- June 2019

Adopted to the integrity pledge for this apex institution, where the Commission is mandated to fight corruption and to ensure integrity in public administration.

NITR International Model United Nations (NITR IMUN)

[Delegate of Bangladesh] 15-17 March 2019

Upheld and presented the role of science and technology in the context of International Security and Disarmament.

NITR Inter Hall Cricket Tournament

[Grand Prize Winner] 2018-19

Champion of the inter-hall cricket tournament at NIT Rourkela.